

Chapter 3

Cartesian Closed Categories

3.1 Exponentials

Definition 3.1.1 (Exponential). Suppose that $\mathcal{C}$ is a category with finite products. An *exponential* of B by A in $\mathcal{C}$ is specified by a pair that consists of:

- An object B^A
- An *evaluation* morphism $ev_{A,B} : B^A \times A \rightarrow B$ such that for any $C \in ob\mathcal{C}$ and $f : C \times A \rightarrow B$, there is a unique morphism $\lambda f : C \rightarrow B^A$ (called the *transpose* or the *exponential mate* of f) such that $f = ev \circ (\lambda f \times id_A)$, i.e. the following diagram commutes.

$$\begin{array}{ccccc}
 B^A & & B^A \times A & \xrightarrow{ev} & B \\
 \uparrow \lambda f & & \uparrow \lambda f \times id_A & & \nearrow f \\
 C & & C \times A & &
 \end{array}$$

Definition 3.1.2 (Cartesian closed category). A Cartesian closed category is a category with finite products such that any pair of objects A and B has an exponential.

Remark 3.1.3. Each $f : C \times A \rightarrow B$ uniquely determines its transpose $\lambda f : C \rightarrow B^A$, by definition, but it is also determined by its transpose, since $f = ev \circ (\lambda f \times id_A)$.

Remark 3.1.4. Every arrow $g : C \rightarrow B^A$ is the transpose of the morphism $\bar{g} \stackrel{def}{=} ev \circ (g \times id_A) : C \times A \rightarrow B$. Note that we also call $\bar{g}$ the transpose of g , while we call λf the transpose of f .

Exercise 3.1.5. Given a Cartesian closed category with arrows $f : C \rightarrow B$ and $h : B \times A \rightarrow T$, show that the transpose of $h \circ (f \times id_A)$ is $\lambda h \circ f$. Conclude that for any $k : B \rightarrow T^A$ the transpose of $k \circ f$ is $\bar{k} \circ (f \times id_A)$.

Exercise 3.1.6. Show that, if (B^A, ev) is an exponential of B by A , then $ev = \bar{id_{B^A}}$.

Exercise 3.1.7. Show that, if (I, ev_i) and (J, ev_j) are both exponentials of B by A , then $I \cong J$.

Exercise 3.1.8. Show that the following properties hold in a Cartesian closed category.

1. $(B + C) \times A \cong (B \times A) + (C \times A)$
2. $(B \times C)^A \cong B^A \times C^A$
3. $C^{(A \times B)} \cong (C^B)^A \cong (C^A)^B$
4. $B^{(A+C)} \cong B^A \times B^C$

3.2 Internalizing Composition

Definition 3.2.1. In a Cartesian closed category, the arrow $f^A : B^A \rightarrow C^A$, the *exponentiation of an arrow* $f : B \rightarrow C$, is defined to be the transpose of $f \circ ev : B^A \times A \rightarrow C$.

Exercise 3.2.2. Show that f^A is a unique arrow that makes the following square commute:

$$\begin{array}{ccccc}
C^A & & C^A \times A & \xrightarrow{ev} & C \\
f^A \uparrow & & f^A \times id_A \uparrow & & \uparrow f \\
B^A & & B^A \times A & \xrightarrow{ev} & B
\end{array}$$

Exercise 3.2.3. Show that $(id_B)^A = id_{B^A}$ and $(g \circ f)^A = g^A \circ f^A$.

Exercise 3.2.4. Given a Cartesian closed category with arrows $f : C \rightarrow T$ and $h : B \rightarrow C^A$, show that the transpose of $f^A \circ h$ is $f \circ \bar{h}$. Conclude that for any $k : B \times A \rightarrow C$ the transpose of $f \circ k$ is $f^A \circ \lambda k$.

Exercise 3.2.5. Show, for any object A and monic $f : B \rightarrow C$, that f^A is monic.

3.3 Strict Initial Object

Definition 3.3.1 (Strict initial object). In any category, an initial object is *strict* if and only if every object with a morphism to it is also initial.

Initial objects in a Cartesian closed category are strict. This can be formally stated as the following theorem.

Theorem 3.3.2. In a Cartesian closed category with an initial object $\emptyset$, if an object A has an arrow $f : A \rightarrow \emptyset$, then $A \cong \emptyset$.

Proof. See the rest of this section. □

Theorem 3.3.3. In a Cartesian closed category with an initial object $\emptyset$, $\emptyset \times A \cong \emptyset$ for every object A .

Proof. For any object B and any arrows $f, g : \emptyset \times A \rightarrow B$, the following diagrams commute.

$$\begin{array}{ccc}
B^A \times A & \xrightarrow{ev} & B \\
\lambda f \times id_A \uparrow & \nearrow f & \\
\emptyset \times A & &
\end{array}
\qquad
\begin{array}{ccc}
B^A \times A & \xrightarrow{ev} & B \\
\lambda g \times id_A \uparrow & \nearrow g & \\
\emptyset \times A & &
\end{array}$$

However, since an arrow from $\emptyset$ to B is unique, $\lambda f = \lambda g$. Then,

$$\begin{aligned} f &= ev \circ (\lambda f \times id_A) \\ &= ev \circ (\lambda g \times id_A) \\ &= g \end{aligned}$$

Hence there is only one arrow from $\emptyset \times A$ to B : that is, $\emptyset \times A$ is an initial object. By Theorem 2.1.5, $\emptyset \times A \cong \emptyset$. $\square$

Remark 3.3.4. By Theorem 2.2.4 and Theorem 1.4.4, we also have $(A \times \emptyset) \cong (\emptyset \times A) \cong \emptyset$.

Theorem 3.3.5. In a Cartesian closed category with an initial object $\emptyset$, $A^\emptyset \cong 1$ for every object A .

Proof. For any object B and any arrows $f, g : B \rightarrow A^\emptyset$, the following diagrams commute.

$$\begin{array}{ccc} A^\emptyset \times \emptyset & \xrightarrow{ev} & A \\ f \times id_\emptyset \uparrow & \nearrow \bar{f} = ev \circ (\lambda f \times id_\emptyset) & \\ B \times \emptyset & & \end{array} \quad \begin{array}{ccc} A^\emptyset \times \emptyset & \xrightarrow{ev} & A \\ g \times id_\emptyset \uparrow & \nearrow \bar{g} = ev \circ (\lambda g \times id_\emptyset) & \\ B \times \emptyset & & \end{array}$$

By Theorem 3.3.3, $B \times \emptyset$ is an initial object and there is only one arrow from $B \times \emptyset$ to A , namely, $\bar{f} = \bar{g}$. Then, $f = \lambda(\bar{f}) = \lambda(\bar{g}) = g$. This means that there is only one arrow from B to $A^\emptyset$, that is, $A^\emptyset$ is a terminal object. By Theorem 2.1.2, $A^\emptyset \cong 1$. $\square$

Proof of Theorem 3.3.2. By Theorem 2.4.6 and Theorem 2.4.2, the graph of $f : A \rightarrow \emptyset$ is monic.

$$\begin{array}{ccccc} & & A & & \\ id_A \swarrow & & \downarrow \Gamma_f & \searrow f & \\ A & \xleftarrow{\pi_1} & A \times \emptyset & \xrightarrow{\pi_2} & \emptyset \end{array}$$

By Theorem 3.3.3, Theorem 1.5.8 and Theorem 1.5.2, there is a monic $m : A \rightarrow \emptyset$.

$$\begin{array}{ccc}
 A & & \\
 \downarrow \Gamma_f & \searrow m & \\
 A \times \emptyset & \xrightarrow{\sim} & \emptyset
 \end{array}$$

On the other hand, by Theorem 2.1.6, any arrow from A to $\emptyset$ is split epic. Therefore, by Theorem 1.5.8, m is iso and $A \cong \emptyset$. $\square$

3.4 Partially-ordered Set

Definition 3.4.1 (Partially-ordered set). A *partially-ordered set* $\langle L; \leq \rangle$ is a set L with the binary relation $\leq$ which is reflexive, anti-symmetric, and transitive.

Definition 3.4.2 (Poset). A *poset* is a category in which, for any objects A and B , there is at most one arrow from A to B , and all the diagrams commute.

Theorem 3.4.3. A *poset* can be construed as a partially-ordered set L , whose elements are the objects of the poset, and for all $x, y \in L$, $x \leq y$ in L iff the poset has an arrow from x to y , and $x = y$ in L iff $x \cong y$.

Proof.

Reflexivity: For any object A , there exists id_A , thus $A \leq A$.

Anti-symmetry: For any object A and B with arrows $f : A \rightarrow B$ and $g : B \rightarrow A$, $g \circ f = id_A$ since the arrow from A to A is unique, and $f \circ g = id_B$ since the arrow from B to B is unique. Thus $A \cong B$. Therefore, if $A \leq B$ and $B \leq A$, then $A = B$.

Transitivity: For any object A , B , and C with arrows $f : A \rightarrow B$ and $g : B \rightarrow C$, there is a composite $g \circ f : A \rightarrow C$. Therefore, if $A \leq B$ and $B \leq C$, then $A \leq C$.

$\square$

Remark 3.4.4. Note that $A \leq B$ iff $A \times B \cong A$ (or $A + B \cong B$) in a poset.

Exercise 3.4.5. Show that a poset is a lattice iff it has binary products and coproducts. (Hint: products and coproducts are meets and joins in a poset.)

Exercise 3.4.6. Show that a poset is a bounded lattice iff it has a terminal object and an initial object. (Hint: a terminal object and an initial object are the top and the bottom elements.)

Exercise 3.4.7. Show that a poset is a Heyting lattice iff it is Cartesian closed. (Hint: exponentials are Heyting implications in a poset.)

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